

Picus 2 pipetting basics over command interface

An electronic pipette, like Picus 2, can be used in multiple ways. In this document you'll find some examples on how to use Picus 2 pipette over Bluetooth or USB command interface in efficient manner.

First some basics. The pipette main motor that we control with commands has several key positions. From top down:

Upper limit: maximum position you can move the piston towards inside, varies between models

Home: Position where you start all pipetting modes.

Blowout: position below home, used to ensure complete dispense of liquid

Tip eject: lowest possible position, actuates tip ejection collar/bar

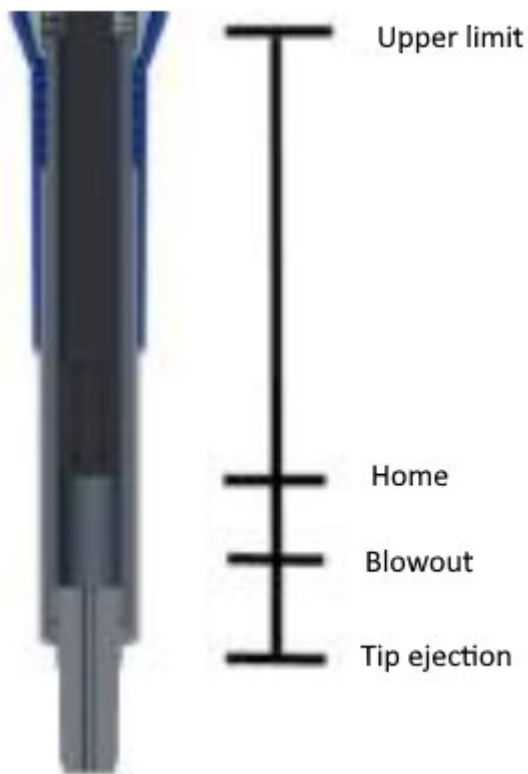


Figure 1Piston positions

Pipetting Modes

Choosing the correct pipetting mode is critical for the pipetting results and it is the first thing to decide before starting pipetting work. With mechanical pipettes there are only two options to choose from (forward and reverse), whereas electronic pipettes provide the user with multiple options to choose from.

Forward pipetting (or just pipetting)

Forward pipetting technique is the default technique in pipetting. Forward pipetting is used when the pipette is factory calibrated and serviced during its lifetime. The specifications for pipettes' systematic and random error are always given for forward pipetting. In forward pipetting the same amount of liquid is aspirated and dispensed.

Forward pipetting consists of an aspiration step, dispensing step, and blow-out. Typically, with electronic pipettes dispensing and blow-out steps are combined. The amount of liquid you wish to transfer is defined in aspiration step; you should aspirate as much as you'd like to deliver. Blow-out is used to ensure that all liquid exits the tip. Without a blow-out step, very often some liquid would remain in tip. After blow-out step it is necessary to return the piston to home position to be ready for next aspiration.

Picus 2 example to pipette 500 μ L, please refer to command listing for more details about the used commands:

1. RUN_ASPIRATE 500 7
2. BLOW_OUT 1 7 3000

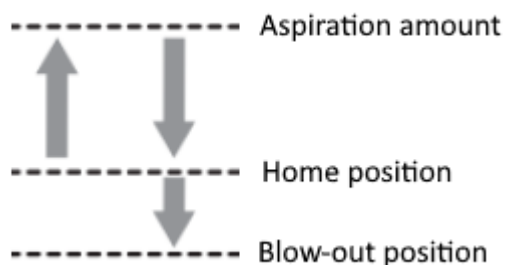


Figure 2 Aspiration, dispensing, and blow-out steps in forward pipetting

Reverse pipetting

Reverse pipetting is recommended for highly viscous, biological, or foaming liquids, or very small volumes. The selected volume is aspirated into the tip along with an excess volume. Delivery is done without blow-out ensuring the excess volume remains in the tip. This excess is then discarded after the selected volume is dispensed.

In reverse pipetting mode the desired volume is achieved in first dispense command. In aspiration step the aim is to aspirate more than the desired volume. Idea behind this is alleviate any potential error in delivery that might come from liquid properties. Reverse pipetting is especially useful for high viscous liquids and when trying to prevent liquid from foaming. Typically for challenging liquids you also should use slower speeds.

Picus 2 example to reverse pipette 500 μ L, please refer to command listing for more details about the used commands:

1. RUN_ASPIRATE 540 4
2. RUN_DISPENSE 500 4
3. BLOW_OUT 1 4 3000

In this example 40 μ L volume was used as excess and was discarded. Actual excess volume can vary based on the liquid used.

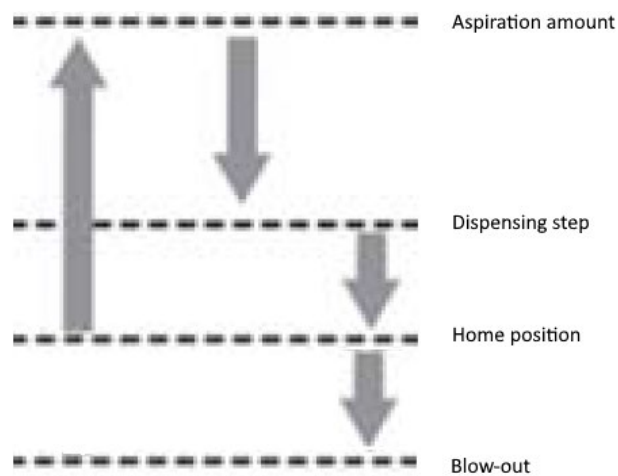


Figure 3 Aspiration, dispensing, and discarding steps in reverse pipetting

Multi-Dispensing

Multi-dispensing aspirates the total volume as well as an excess volume, then repetitively dispenses equal volumes of liquid. It is recommended for long pipetting series and microplate dispensing. This function is only available in electronic pipettes.

Also it's worth noting that depending on the use case, it can be beneficial to use different size aliquots for dispensing.

In multi-dispensing mode you first aspirate the full amount, including excess amount. For pipetting operations including multiple aliquots (dispensings), it is recommended to evaluate the need for pre-out excess (purple in the picture). This increases the repeatability of 1st dispensed aliquot, especially when used with high viscous liquids.

Picus 2 example to multi-dispense 3 x 100 μ L, please refer to command listing for more details about the used commands:

1. RUN_ASPIRATE 360 4
2. RUN_DISPENSE 30 4

3. RUN_DISPENSE 100 4
4. RUN_DISPENSE 100 4
5. RUN_DISPENSE 100 4
6. BLOW_OUT 1 4 3000

In this example, 30µL of volume was used for pre-out, and 30µL were ejected together with blow-out.

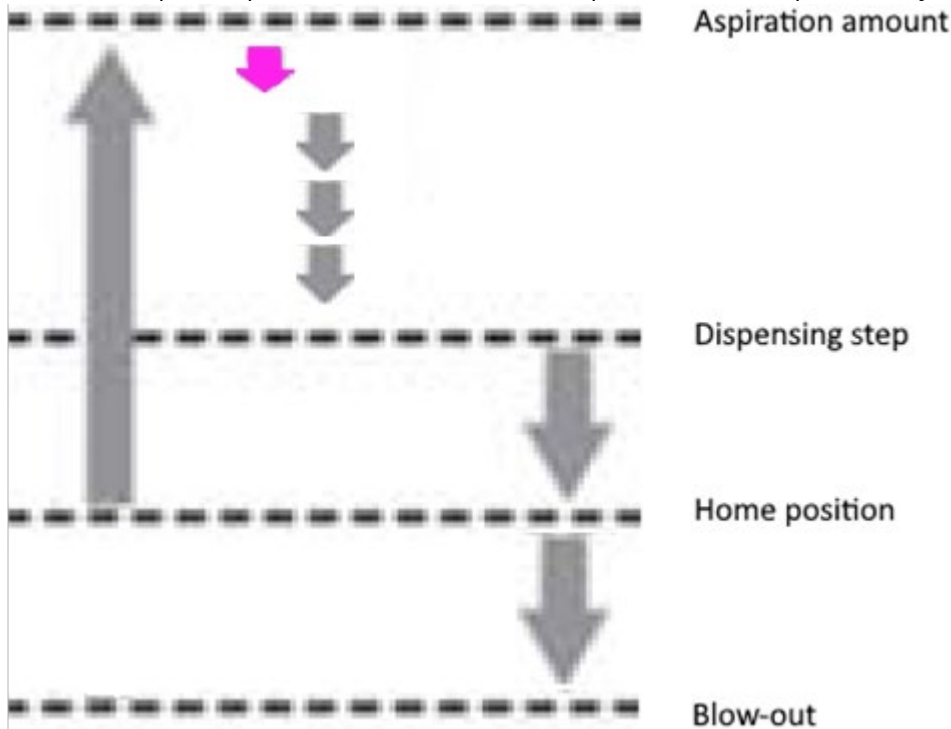


Figure 4 Multi-dispensing steps

Pipette Tips

The most reliable pipetting results are obtained when using a system with a well optimized tip fit. Sartorius Optifit or Safetyspace tips ensure a seamless fit and best possible performance with Sartorius pipettes. All Sartorius tips are manufactured with diamond polished molds, a process that yields a smooth and consistent tip surface, minimizing liquid retention, thereby facilitating precise and accurate pipetting results.

The high-quality standard Optifit tips are the preferred choice for a wide range of laboratory applications and pipetting tasks, while the Safetyspace Filter tips ensure contamination-free operation. The broad selection of tip packaging and purity options allows selecting the best suited product for a specific laboratory setting.